

R&D Tax Credit Eligibility – A Narrative

Burhan Rangwala, The University of Illinois Urbana Champaign

Company

A Texas-based life-sciences software startup providing AI-driven recruitment tools for clinical trials.

Project Overview

The organization builds AI-driven automation tools for clinical research organizations. The company focuses on software that helps sites and sponsors recruit and manage trial participants more efficiently, often by connecting social-media outreach and digital campaigns to internal workflows.

The project under review is the development of a patient-matching and prescreening module that links leads from Facebook, Instagram, and other platforms to a centralized engine that directs them to the correct trial sites.

Project Description

The team is building a system that:

- Generates leads from social-media and digital-marketing campaigns.

- Uses basic rules (not full-fledged AI models) to assign leads to trial sites based on geography, and availability.

- Attempts to automate scheduling of initial prescreening calls and sends basic alerts to site coordinators.

- Connects to platforms and internal CRMs to keep lead status updated.

Work is done by backend engineers, data-science consultants, and product managers. Most documentation is in Jira tickets, and ad-hoc meeting recaps. The project is under progress, and the way work was planned, and document raises questions about true R&D treatment.

Application of the Four-Part Test (Qualitative)

Permitted purpose: The module is part of the firm's clinical-research platform and is intended to improve how leads are routed and prescreened. It changes the existing workflow and adds new automation, so it likely meets the "permitted purpose" element.

Technical uncertainty: Based on my discussion, the following questions arose

How to reliably ingest leads from multiple social-media channels without missing records.

How to design a simple routing logic that avoids overloading certain sites.

How to keep HIPAA-adjacent data clean and secure in a shared environment.

These uncertainties are typical of early-stage software projects and suggest some room for R&D-style thinking.

Process of experimentation – where the project falls short

This is where the project fails the R&D test.

The team did not really follow a true process of experimentation; instead:

They picked a single routing logic upfront and stuck with it, even when early data showed uneven site load and some sites complaining about lead volume.

They built only one version of the automation engine and deployed it without meaningful A/B tests or alternative designs.

They did not document tests, failure modes, or design iterations that would show learning-based development; most notes are 'task done' style, not 'experiment X failed, so we changed Y.'

In R&D-tax terms, there is movement and change, but not systematic experimentation. The work was more like a straightforward software build than a discovery-driven R&D project.

Reliance on principles of the hard sciences: Some of the work relies on computer-science and software-engineering principles (API design, workflow automation, basic rule-based matching). However, the lack of experimentation would mean the project is more like applied engineering than true R&D under the tax rules.

Documentation Quality and Risk

Project notes are scattered and not written in a way that would show a learning-based approach. There are no clear records of:

Prototype versions,

Failed routing strategies,

Performance- vs. fairness-trade-off decisions.

Because the team did not document experiments, it would be hard to defend the “process of experimentation” element in an IRS review.

The project also mixes R&D-looking work with routine integration and configuration, which only increases the risk of disallowance.

Conclusion and Recommendation

Qualitatively, this project likely fails the R&D tax credit test because it does not meet the process of experimentation requirement. The team did face some technical uncertainty and built a new feature, but they did not systematically test alternatives, iterate based on results, or document those experiments in a way that would support a credit claim.

That said, the project has been instrumental as a learning example:

It shows how a software project in clinical research can look like R&D but still fall short on the experimentation element.

It highlights the importance of documenting tests, failures, and design choices, not just ‘completed tasks.’

For future engagements, the startup can:

Design projects with clear experimentation in mind (e.g. multiple prototypes, A/B tests, documented trade-offs).

Capture those experiments in short, plain-language narratives tied to Form 6765